Day 16 Of Communication System Class

6 Baseband Digital Data Transmission (8 hours)
6.1 Information theory, measurement of information, entropy, symbol rates and data rates
6.2 Shannon Hartley channel capacity theorem, implication of theorem, and theoretical limits
6.3 Compression techniques: Shannon-Fano, Huffman codes

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6.1 Information theory, measurement of information, entropy, symbol rates and data rates

1. Define information and entropy. [Most Probable Question]

Information is the amount of uncertainty reduced when a message or event is received. The less likely an event is, the more information it conveys.

Entropy is the average information content or the average uncertainty of a source. It represents the average number of bits required to describe the output of a source.

$$H(X) = -\sum_{x} p(x) \log p(x)$$

2. Differentiate between message and information .[Most Probable Question]

3. What is the aim of source coding? [Most Probable Question]

Source coding is the process of representing information using the minimum number of bits without losing essential information (or with acceptable loss in lossy compression).

Objectives:

- Reduce redundancy in the source data.
- Compress data to save storage and transmission bandwidth.
- Increase transmission efficiency.
- Reduce transmission time and cost.
- Improve the overall efficiency of communication systems.

Examples of source coding techniques:

- Huffman Coding
- Shannon-Fano Coding

6.2 Shannon Hartley channel capacity theorem, implication of theorem, and theoretical limits

The Shannon-Hartley Theorem expresses the channel capacity C as:

$C = B\log_2\left(1 + \frac{S}{N}\right)$

Where:

- C: Channel capacity (in bits per second, bps), the maximum error-free data rate.
- B: Bandwidth of the channel (in hertz, Hz), the range of frequencies the channel can transmit.
- S: Signal power (in watts), the strength of the transmitted signal.
- N: Noise power (in watts), the strength of the unwanted interference or noise in the channel.
- $\frac{S}{N}$: Signal-to-noise ratio (SNR), a dimensionless ratio indicating how much stronger the signal is compared to the noise.
- $\log_2$: Logarithm base 2, used because information is measured in bits (binary digits, 0 or 1).

Implications of the Theorem

1. Error-Free Transmission is Possible:
2. Bandwidth vs. SNR Trade-off:
3. The Hard Ceiling (B > C)

Theoretical Limits

1. The Infinite Bandwidth Limit
   If we take the mathematical limit of the Shannon equation as B to infty, the capacity approaches a strict maximum bound:

6.3 Compression techniques: Shannon-Fano, Huffman codes

6.3.1 Shannon-Fano

Shannon-Fano encoding is the first established and widely used encoding method. This method and the corresponding code were invented simultaneously and independently of each other by C. Shannon and R. Fano in 1948.

The Shannon-Fano Encoding Algorithm

1. Calculate the frequencies of the list of symbols (organize as a list).
2. Sort the list in (decreasing) order of frequencies.
3. Divide list into two halfs, with the total freq. *Counts of each half being as close as possible to the other.*
4. The upper half is assigned a code of 0 and lower a code of 1.
5. Recursively apply steps 3 and 4 to each of the halves, until each symbol has become a corresponding code leaf on a tree.

Example

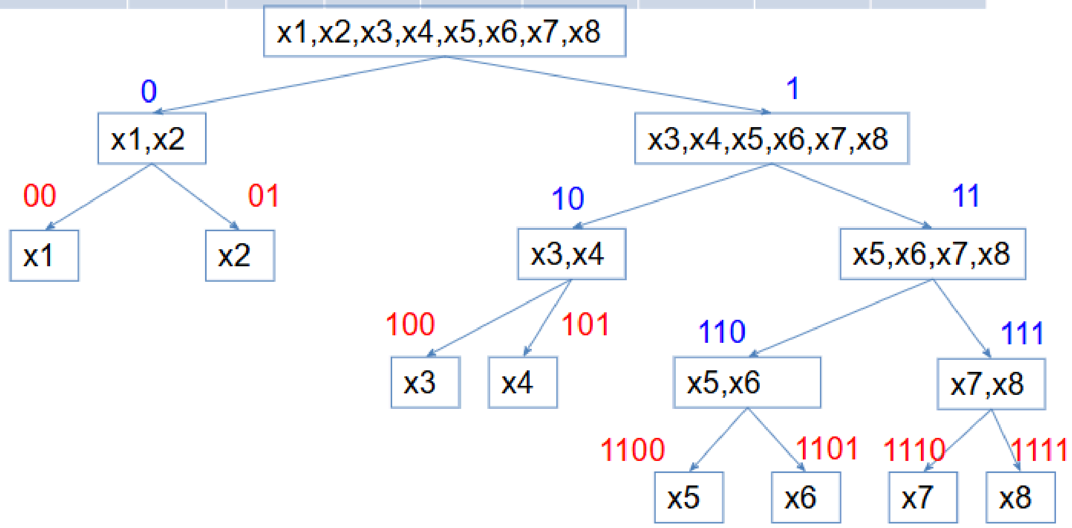
Symbol	A	B	C	D	E
Count	15	7	6	6	5
	0	0	1	1	1
	0	1	0	1	1
				0	1

Symbol	Count		Info. $-\log_2(p_i)$	Code	Subtotal # of Bits
A	15	x	1.38	00	30
B	7	x	2.48	01	14
C	6	x	2.70	10	12
D	6	x	2.70	110	18
E	5	x	2.96	111	15
	85.25				89

It takes a total of 89 bits to encode 85.25 bits of information (Pretty good huh!)

Q. Explain Shannon-Fanocoding. A discrete memoryless source has an alphabet 8 symbols whose probabilities occurrence are as under. Determine the Shannon-Fano for above source.

Message	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8
Probability	0.25	0.25	0.125	0.125	0.0625	0.0625	0.0625	0.0625



- Entropy $H = -\left(2\cdot\left(\frac{1}{4}\log\frac{1}{4}\right) + 2\cdot\left(\frac{1}{8}\log\frac{1}{8}\right) + 4\cdot\left(\frac{1}{16}\log\frac{1}{16}\right)\right) = 2.75$
- Average length of the encoding vector

$$\bar{L} = \sum P\{x_i\} n_i = \left(2\cdot\left(\frac{1}{4}\cdot 2\right) + 2\cdot\left(\frac{1}{8}\cdot 3\right) + 4\cdot\left(\frac{1}{16}\cdot 4\right)\right) = 2.75$$

- The Shannon-Fano code gives 100% efficiency

6.3.2 Huffman codes

Invented by Huffman as a class assignment in 1950.

Used in many, if not most, compression algorithms

- gzip, bzip, jpeg (as option), fax compression,…

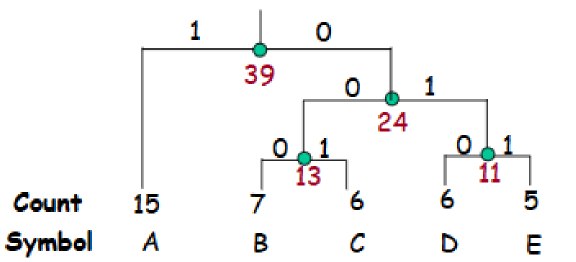
Properties:

- Generates optimal prefix codes
- Cheap to generate codes
- Cheap to encode and decode
- $l_a = H$ if probabilities are powers of 2

The Huffman Algorithm

1. Initialization: Put all nodes in an OPEN list *L*, keep it sorted at all times (e.g., ABCDE).
2. Repeat the following steps until the list *L* has only one node left:
    1. From *L* pick two nodes having the lowest frequencies, create a parent node of them.
    2. Assign the sum of the children's frequencies to the parent node and insert it into L.
    3. Assign code 0, 1 to the two branches of the tree, and delete the children from L.

Example



Symbol	Count		Info. $-\log_2(p_i)$	Code	Subtotal # of Bits
A	15	x	1.38	1	15
B	7	x	2.48	000	21
C	6	x	2.70	001	18
D	6	x	2.70	010	18
E	5	x	2.96	011	15
	85.25				87

Encoding: Start at leaf of Huffman tree and follow path to the root. Reverse order of bits and send.

Decoding: Start at root of Huffman tree and take branch for each bit received. When at leaf can output message and return to root.